# AN EXHAUSTIVE SEARCH FOR FOUR-PRIME WILLIAMS NUMBERS (CARMICHAEL NUMBERS THAT ARE ALSO LUCAS–CARMICHAEL) TO $10^{30}$, AND A VERIFIED BOUND OF $10^{24}$ FOR ALL OF THEM

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ABSTRACT. A squarefree composite $n$ is *Carmichael* if $(p-1) \mid (n-1)$ for every prime $p \mid n$, and *Lucas–Carmichael* if $(p+1) \mid (n+1)$ for every $p \mid n$. Numbers that are both—*Williams numbers*, after Williams's 1977 question—are not known to exist; they are the natural candidates for a Baillie–PSW counterexample (Pomerance) and for the Lenstra–Pomerance route to a counterexample to Agrawal's conjecture. Williams proved that such a number cannot have three prime factors, and McIntosh and Dipra (2015) proved that, with all but three prime factors fixed, only finitely many Williams numbers exist, with explicit bounds: for four prime factors $p_1 < p_2 < p_3 < p_4$, all $p_i \equiv 1 \pmod{12}$, $p_2 < 5p_1^{3/2}$, $p_3 < p_1p_2$ and $p_4 < \sqrt{2p_1p_2p_3}$; they showed that a four-prime example must have $p_1 > 2 \times 10^4$ and $n > 10^{24}$. We give a self-contained proof of these bounds (with $p_2^2 \le 24p_1^3 + 1$) and a different search algorithm, which fixes $(p_1, p_2)$, places the cofactor $p_3p_4$ in a single residue class modulo $\operatorname{lcm}\big((p_1^2-1)/2, (p_2^2-1)/2\big)$ and rejects almost every candidate with one multiplication; with it we extend the exclusion to $n \le 10^{30}$ and to all $n$ with $p_1 \le 500{,}000$, with a certified completeness record. We also verify, by an independent factor-and-test pass over the complete tabulations, that no Carmichael number below $10^{24}$ is Lucas–Carmichael (308,279,939 numbers; at most four of the primes of any of them satisfy $(p+1) \mid (n+1)$). In appendices we show that the two classical construction paradigms for Carmichael numbers fail in their simplest two-sided forms (no Chernick-type polynomial family of three factors, of any number of factors with uniform offsets $\pm 1$, or in generic position can produce such numbers; the two-sided subset-product method is measured to lose about 1.5 bits per prime), and we record heuristic estimates.

## 1. INTRODUCTION

Throughout, $n$ denotes an odd squarefree composite integer with prime factorisation $n = p_1p_2\cdots p_k$, $p_1 < \cdots < p_k$, $k = \omega(n) \ge 2$.

**Definition 1.1.** $n$ is a *Carmichael number* if $(p-1) \mid (n-1)$ for every $p \mid n$ (Korselt's criterion), and a *Lucas–Carmichael number* if $(p+1) \mid (n+1)$ for every $p \mid n$. A number that is both is a *Williams number* (a 1-Williams number in Echi's terminology [9]).

Carmichael numbers are the absolute Fermat pseudoprimes; Lucas–Carmichael numbers play the same role for Lucas sequences with discriminant a quadratic non-residue of every $p \mid n$. H. C. Williams [27] asked in 1977 whether a number can be both, proved that it cannot have exactly three prime factors and cannot be divisible by 3, and showed that under his Lucas-sequence formulation the number of prime factors must be odd (in the bare divisibility formulation above, all $p_i$ lie in one class modulo 4, and $k$ is odd when that class is 3; see Lemma 2.2). Echi [9] named such numbers *Williams numbers* (more precisely 1-Williams numbers, within a family of $a$-Williams numbers) and dropped the parity restriction; Bouallègue, Echi and Pinch [4] and Al-Rasasi and Ghanmi [1] studied the general $a$; Pinch [20] reported that there is no Williams number up to $10^{16}$ with an odd number of prime factors. McIntosh [17] treated the parallel problem $(p \pm 1) \mid (n-1)$ and, with Dipra [18], the Williams numbers themselves: they proved

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that for a fixed set of $d-3$ prime factors only finitely many Williams numbers with $d$ prime factors exist, with explicit bounds on the three remaining primes, observed that all prime factors are congruent modulo 12 (and $\equiv 1$ when their number is even), organised searches, and showed that a Williams number with exactly four prime factors has smallest prime $p_1 > 2 \times 10^4$ and $n > 10^{24}$; they also report that a Pinch-style complete search for $4 \le d \le 13$ prime factors, at various bounds between $10^{20}$ and $10^{35}$, found none. Zhang [26] developed counting heuristics for Williams numbers built from a fixed set of primes, exhibited a set of 1,029 primes expected to contain many of them, and offered a prize for the first example. No Williams number is known.

The question is tied to primality testing. Pomerance's 1984 heuristic argument that counterexamples to the Baillie–PSW primality test exist [22], restated in [5], builds $n = \prod p$ from primes with $p \equiv 3 \pmod 8$, $(5/p) = -1$ and smooth squarefree $(p-1)/2$, $(p+1)/4$, chosen so that $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for all $p \mid n$; such an $n$ is a Williams number, a connection made explicit in [17, 18, 26]. The construction attempted by Chen and Greene [6] for Grantham's \$620 problem [13] uses the weaker order conditions $\mathrm{ord}_p(2) \mid M$; the related set of 2,030 primes they discuss does satisfy $(p-1) \mid M$, $(p+1) \mid N$, and an unrefereed report by Phatak [19] states that no subset of that set is a Williams number. The Lenstra–Pomerance analysis of Agrawal's conjecture from the AKS primality test [2, 16] shows that the natural counterexamples are Williams numbers whose primes are all $\equiv 3 \pmod{80}$. Howe's higher-order Carmichael numbers [14] are a further neighbour: by Howe's criterion, a squarefree $n$ is Carmichael of order 2 iff for every $p \mid n$ one has $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$, while $n$ is a Williams number iff $n \equiv p \pmod{(p^2-1)/2}$ for every $p \mid n$ (Lemma 2.1). We use the name "Williams number" throughout; the abbreviation CLC (Carmichael and Lucas–Carmichael) in the code and the earlier notes means the same thing.

Our results are the following.

- **Four prime factors** (Theorems 3.1 and 3.8). Theorem 3.1 restates, with a self-contained proof, the bounds of McIntosh and Dipra [18, Thms. 2–3] for $d = 4$: all $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$ (they have $p_3 < p_1p_2$), $p_4 < \sqrt{2p_1p_2p_3}$, and $p_2^2 \le 24p_1^3+1$ (they have $p_2 < 5p_1^{3/2}$); so $n < 477\,p_1^{15/2}$ and for each $p_1$ a finite search remains. Theorem 3.8 is new: by a different algorithm (§3.2) we show that no four-prime Williams number has $n \le 10^{30}$, and that none exists with $p_1 \le 500{,}000$, extending their $n > 10^{24}$, $p_1 > 2 \times 10^4$.
- **Verified bound** (Theorem 4.1). No Williams number of any number of prime factors is smaller than $10^{24}$. This is a complete factor-and-test pass over the tabulations of all Carmichael numbers below $10^{24}$, with completeness and per-class cross-checks against independently published counts. The largest number of primes $p \mid n$ with $(p+1) \mid (n+1)$ attained below $10^{24}$ is four (first at a number with $\omega(n) = 9$); 826 Carmichael numbers attain exactly three.
- **Heuristics** (§5, speculative). A simple probabilistic model puts the expected number of four-prime Williams numbers near $10^{-6}$, concentrated at $p_1 = 13$.
- **Appendices.** Appendix A shows that the two classical constructions of Carmichael numbers fail in their simplest two-sided forms: Chernick-type polynomial families cannot produce Williams numbers with three factors, with uniform offsets $\pm 1$, or in generic position (Theorems A.1–A.3), the degenerate strata are empty in the ranges searched, and the two-sided subset-product method is measured to lose about 1.5 bits per prime. Appendix B gives a heuristic onset estimate.

All code, logs and result files are available (§6), together with a certification record for the enumerations.

## 2. Preliminaries

For an odd prime $p$ write $L_p = (p^2-1)/2 = \operatorname{lcm}(p-1, p+1)$.

**Lemma 2.1.** *Let $p \mid n$. Then $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ hold together if and only if $n/p \equiv 1 \pmod{L_p}$, equivalently $n \equiv p \pmod{L_p}$.*

*Proof.* The two divisibilities say $n \equiv 1 \pmod{p-1}$ and $n \equiv -1 \pmod{p+1}$. Since $p \equiv 1 \pmod{p-1}$ and $p \equiv -1 \pmod{p+1}$, this is $n \equiv p$ modulo $\operatorname{lcm}(p-1, p+1) = L_p$ (the gcd of $p \mp 1$ is 2). Finally $L_p \mid n - p = p\,(n/p - 1)$ with $\gcd(p, L_p) = 1$ gives $L_p \mid n/p - 1$, and conversely. □

**Lemma 2.2** (congruence classes)**.** *Let $n$ be a Williams number with $k$ prime factors.*

(a) *All $p \mid n$ lie in one class $c$ modulo 4, and $n \equiv c \pmod 4$; hence $k$ is odd if $c = 3$.*
(b) *$3 \nmid n$; all $p \mid n$ lie in one class $c'$ modulo 3, and $n \equiv c' \pmod 3$; hence $k$ is odd if $c' = 2$.*
(c) *If $k$ is even, every $p \mid n$ is $\equiv 1 \pmod{12}$.*

*Proof.* (a) $8 \mid p^2 - 1$ for odd $p$, so $4 \mid L_p$ and $n \equiv p \pmod 4$ for every $p \mid n$ by Lemma 2.1. Thus all $p \equiv c$, and $n \equiv c^k \pmod 4$; if $c = 3$ this forces $k$ odd. (b) For $p \neq 3$, $3 \mid p^2 - 1$, so $3 \mid L_p$ and $n \equiv p \pmod 3$. If $3 \mid n$ then every other prime factor would be $\equiv n \equiv 0 \pmod 3$, impossible for $k \geq 2$; so $3 \nmid n$, all $p \equiv c' \pmod 3$ and $n \equiv c'^k \equiv c'$, which for $c' = 2$ forces $k$ odd. (c) is (a) and (b) with $k$ even. □

Lemma 2.2(c) is observed in [18, §2]. Williams proved that $k = 3$ is impossible; the following short proof is essentially that of [18, Thm. 1].

**Proposition 2.3** (Williams)**.** *No Williams number has exactly three prime factors.*

*Proof.* Let $n = p_1p_2p_3$. By Lemma 2.1 at $p_3$, $p_1p_2 - 1 = tL_3$ with $t \geq 1$ (as $p_1p_2 > 1$); since $p_1p_2 < p_3^2$ we have $t < 2$, so $2p_1p_2 = p_3^2 + 1$. At $p_2$, $p_1p_3 - 1 = sL_2$ with $s \geq 1$; now $p_3 < \sqrt{2p_1p_2} < \sqrt{2}\,p_2$, so $p_1p_3 < \sqrt{2}\,p_2^2$ and $s < 2\sqrt{2}\,p_2^2/(p_2^2-1) < 3$ (as $p_2 \geq 5$). If $s = 2$ then $p_1p_3 = p_2^2$, impossible; so $2p_1p_3 = p_2^2 + 1$. Subtracting, $2p_1(p_2 - p_3) = p_3^2 - p_2^2$, i.e. $2p_1 = -(p_2 + p_3) < 0$, a contradiction. □

## 3. Four prime factors

### 3.1. Structure.

**Theorem 3.1** (McIntosh–Dipra [18], $d = 4$; restated)**.** *Let $n = p_1p_2p_3p_4$, $p_1 < p_2 < p_3 < p_4$, be a Williams number, and write $P = p_1p_2$, $q = p_3$, $r = p_4$.*

(a) *Every $p_i \equiv 1 \pmod{12}$; in particular $p_1 \geq 13$ and $p_2 \geq 37$.*
(b) *There are integers $d, e, s \geq 1$ with*

$$2Pq = d(r^2-1) + 2, \qquad 2Pr = e(q^2-1) + 2, \qquad 2p_1qr = s(p_2^2-1) + 2, \tag{1}$$

*and consequently $q < 2P$, $r < \sqrt{2Pq} < 2P$, $m := qr < 4P^2$, $n < 4P^3$ and $d < 2p_1$.*
(c) *$p_2^2 \leq 24p_1^3 + 1$.*

*Consequently $p_2 < 4.9\,p_1^{3/2}$, $p_3, p_4 < 2P < 9.8\,p_1^{5/2}$ and $n < 477\,p_1^{15/2}$.*

*Proof.* (a) is Lemma 2.2(c).

(b) Lemma 2.1 at $r$ gives $n/r - 1 = Pq - 1 = dL_r$ with an integer $d \geq 1$, which is the first equation; the second and third are Lemma 2.1 at $q$ and at $p_2$. From $2Pq = d(r^2-1)+2 \geq r^2+1 > q^2$ we get $q < 2P$ and $r^2 < 2Pq$, hence $r < \sqrt{2Pq} < 2P$, $m = qr < 4P^2$ and $n = Pm < 4P^3$. Also $d = 2(Pq-1)/(r^2-1) < 2Pq/(r^2-1) < 2P/q < 2p_1$ because $r \geq q+1$ and $q > p_2$.

(c) Multiply the three equations in (1). The left side is $8P^2p_1\,q^2r^2 = 8p_1^3p_2^2q^2r^2$. Writing $d(r^2-1)+2 = dr^2\big(1-\frac{d-2}{dr^2}\big)$ and similarly for the other two factors, the right side is $d\,e\,s\;r^2q^2p_2^2\,\Pi$ with

$$\Pi = \Big(1-\frac{d-2}{dr^2}\Big)\Big(1-\frac{e-2}{eq^2}\Big)\Big(1-\frac{s-2}{sp_2^2}\Big).$$

Hence

$$d\,e\,s\cdot\Pi = 8p_1^3. \tag{2}$$

For an integer $t\ge 1$ and real $x>0$ one has $-1/x^2 \le (t-2)/(tx^2) < 1/x^2$, so each factor of $\Pi$ lies in $[1-1/x^2,\,1+1/x^2]$ with $x\in\{r,q,p_2\}$, $x\ge p_2$; therefore

$$(1-p_2^{-2})^3 \le \Pi \le (1+p_2^{-2})^3. \tag{3}$$

Now reduce the three equations (1) modulo $p_1$. Each left side is divisible by $p_1$, so $d(r^2-1)\equiv e(q^2-1)\equiv s(p_2^2-1)\equiv -2 \pmod{p_1}$. As $p_1$ is odd, $p_1\nmid d$, $p_1\nmid e$ and $p_1\nmid s$. Hence $des\neq 8p_1^3$, i.e. $\Pi\neq 1$.

*Case* $\Pi<1$. Then $des\ge 8p_1^3+1$ and $\Pi = 8p_1^3/des \le 1-\frac{1}{8p_1^3+1}$; with $\Pi\ge(1-p_2^{-2})^3\ge 1-3p_2^{-2}$ this gives $p_2^2\le 3(8p_1^3+1) = 24p_1^3+3$. Since $p_2\equiv 1 \pmod{12}$ implies $p_2^2\equiv 1 \pmod{24}$, while $24p_1^3+2$ and $24p_1^3+3$ are $\equiv 2,3 \pmod{24}$, in fact $p_2^2\le 24p_1^3+1$.

*Case* $\Pi>1$. A factor $1-\frac{t-2}{tx^2}$ of $\Pi$ exceeds 1 only when $t=1$. Now $e>d\ge 1$, because $e/d = \frac{(Pr-1)(r^2-1)}{(Pq-1)(q^2-1)} > 1$ for $r>q$; and $s>2p_1$, because $qr\ge p_2^2+1$ gives $s = \frac{2(p_1qr-1)}{p_2^2-1} \ge \frac{2(p_1p_2^2+p_1-1)}{p_2^2-1} > 2p_1$. So only the first factor can exceed 1, and $\Pi\le 1+r^{-2}$. On the other hand $des\le 8p_1^3-1$ gives $\Pi = 8p_1^3/des \ge 1+\frac{1}{8p_1^3-1}$. Hence $r^2\le 8p_1^3-1$, and $p_2^2<r^2<8p_1^3\le 24p_1^3+1$.

In both cases $p_2^2\le 24p_1^3+1$.

The consequences follow from (b) and (c): $p_2^2\le 24p_1^3+1 < 24.01\,p_1^3 = (4.9\,p_1^{3/2})^2$ for $p_1\ge 5$, so $p_2<4.9\,p_1^{3/2}$; then $q,r<2P<9.8\,p_1^{5/2}$ and $n<4P^3<4\cdot 4.9^3\,p_1^{15/2}<477\,p_1^{15/2}$. □

*Remark* 3.2. An earlier draft handled the case $\Pi>1$ with the estimate $(1+p_2^{-2})^3 \le 1+3.0001\,p_2^{-2}$, which is false (the correct coefficient at $p_2=37$ is 3.0022, and with it that route does not close); the error was found in the independent check described in §6. The enumeration of §3.2 used the bound $p_2^2\le 24p_1^3+3$, which both cases satisfy, so Theorem 3.8 is unaffected.

**Corollary 3.3.** *For each prime $p_1$ there are only finitely many four-prime Williams numbers with smallest prime $p_1$, all below $477\,p_1^{15/2}$. Conversely a four-prime Williams number $n$ has $p_1>(n/477)^{2/15}$.*

*Remark* 3.4 (relation to McIntosh and Dipra). Theorem 3.1 is Theorems 2 and 3 of [18] specialised to $d=4$ (their $P,q,r,s$ are our $p_1,p_2,p_3,p_4$, their $f_s,f_r,f_q$ our $d,e,s$): they prove $q<5P^{3/2}$, $r<Pq$ and $s<\sqrt{2Pqr}$, and show directly, using Newton's inequality $S_2^2\ge S_1S_3$, that the quotient product always exceeds $8P^3$ (so $\Pi<1$ and the second case in our proof never occurs). Our statement differs only in the trivially sharper $p_2^2\le 24p_1^3+1$ (from the congruence modulo 24) and the weaker $p_3<2p_1p_2$, which is all the enumeration needs; the proof above is independent but uses the same product identity. We had derived it before learning of [18].

3.2. **Enumeration.** Theorem 3.1 makes two kinds of complete enumeration possible: all $n\le B$ (where $n>p_1p_2^3$ forces $p_1<B^{1/4}$ and $p_2<(B/p_1)^{1/3}$), and all $n$ whatsoever with $p_1\le X$ (where $p_2\le\sqrt{24p_1^3+1}$). Both reduce to a search over pairs $(p_1,p_2)$.

**Lemma 3.5** (one class for the cofactor)**.** *With notation as in Theorem 3.1, $m=qr$ satisfies $p_2m\equiv 1 \pmod{L_{p_1}}$ and $p_1m\equiv 1 \pmod{L_{p_2}}$. These are solvable iff $p_1\nmid p_2^2-1$ and $p_1\equiv p_2 \pmod{\gcd(L_{p_1},L_{p_2})}$, in which case $m$ lies in a single residue class modulo $\Lambda=\operatorname{lcm}(L_{p_1},L_{p_2})$.*

*Proof.* The congruences are Lemma 2.1 at $p_1$ and $p_2$ ($n/p_1 = p_2 m$, $n/p_2 = p_1 m$). The first congruence is always solvable: $p_2 \mid L_{p_1}$ would need $p_2 \mid p_1 \pm 1$ with $p_2 > p_1$, impossible for odd primes. The second is solvable iff $\gcd(p_1, L_{p_2}) = 1$, i.e. $p_1 \nmid p_2^2 - 1$. Given both, let $g = \gcd(L_{p_1}, L_{p_2})$; the pair is compatible iff $p_2^{-1} \equiv p_1^{-1} \pmod g$, i.e. $p_1 \equiv p_2 \pmod g$ (note $\gcd(p_1 p_2, g) = 1$). The rest is the Chinese remainder theorem. □

Since $\Lambda \approx P^2/(4g)$ while $m < 4P^2$, Lemma 3.5 leaves $O(g)$ candidates $m$ per pair. The next lemma rejects almost all of them with one multiplication.

**Lemma 3.6** (window)**.** *With $d, e$ as in (1) and $j := de$, $T := 4P^2$,*

$$|jm - T| < \max\bigl(2j\sqrt{2p_1},\ 8p_1\bigr). \tag{4}$$

*Moreover, for fixed $m > 2\lceil\sqrt{2p_1}\rceil$ (which holds for $m > p_2^2$) the set of real $j > 0$ satisfying $|jm - T| \le \max(2j\lceil\sqrt{2p_1}\rceil, 8p_1)$ is an interval containing $T/m$; hence it contains an integer iff it contains $\lfloor T/m\rfloor$ or $\lceil T/m\rceil$.*

*Proof.* From (1), $d = 2(Pq-1)/(r^2-1)$ and $e = 2(Pr-1)/(q^2-1)$, so $j\,(q^2-1)(r^2-1) = 4(Pq-1)(Pr-1)$. Expanding with $m = qr$,

$$jm^2 - Tm = j(q^2 + r^2 - 1) - 4P(q+r) + 4.$$

The right side is a difference of two positive numbers, so its absolute value is at most $\max\bigl(j(q^2+r^2-1),\ 4P(q+r)-4\bigr) \le \max(2jr^2,\ 8Pr)$. Dividing by $m = qr$ gives $|jm - T| \le \max(2jr/q,\ 8P/q)$. By Theorem 3.1(b), $r^2 < 2Pq$, so $r/q < \sqrt{2P/q} < \sqrt{2p_1}$ (as $q > p_2$), and $P/q < p_1$; this is (4). For the interval property, put $S = \lceil\sqrt{2p_1}\rceil$ and $\phi(j) = |jm - T| - \max(2jS, 8p_1)$. On $j \ge T/m$ the first term increases with slope $m$ and the second with slope at most $2S < m$, so $\phi$ is strictly increasing; on $j \le T/m$ the first term decreases and the second is non-decreasing, so $\phi$ is strictly decreasing. Thus $\{\phi \le 0\}$ is an interval, and it contains $T/m$ since $\phi(T/m) < 0$. □

Algorithm. Put $S = \lceil\sqrt{2p_1}\rceil$, $R_j = \max(2jS, 8p_1)$, and call $(m, j)$ *admissible* if $|jm - T| \le R_j$; by Lemma 3.6 the true $(m, de)$ is admissible, and for fixed $m > p_2^2 > 2S$ the admissible $j$ form an interval containing $T/m$. For every pair of primes $p_1 < p_2$, both $\equiv 1 \pmod{12}$, within the range dictated by the campaign: (i) skip the pair unless both conditions of Lemma 3.5 hold (a pair failing them has no CLC completion), and compute $m_0$, $\Lambda$; (ii) the candidates are $m \equiv m_0 \pmod\Lambda$ with $p_2^2 < m \le m_{\max} := \min(4P^2 - 1, \lfloor B/P\rfloor)$; choose a threshold $M^* \approx \sqrt{T\Lambda}$; for $m \le M^*$ walk the candidates and keep $m$ iff $\lfloor T/m\rfloor$ or $\lceil T/m\rceil$ is admissible for it; for $m > M^*$ walk instead the integers $j$ from $\max\bigl(1, \lfloor(T-8p_1)/(m_{\max}+2S)\rfloor - 1\bigr)$ to $\lfloor(T+8p_1)/(M^*-2S)\rfloor + 1$ and keep every candidate $m \in (M^*, m_{\max}]$ in the exact window $\lceil(T-R_j)/j\rceil \le m \le \lfloor(T+R_j)/j\rfloor$ (any admissible $(m, j)$ with $m > M^*$ has $j$ in that range, since $jm \le T + 2jS + 8p_1$ and $jm \ge T - 2jS - 8p_1$); the two walks were checked to produce identical survivor lists on six instances, including two relaxed instances with 346,409 and 11,007,875 survivors; (iii) for each surviving $m$ require $2^{n-1} \equiv 1 \pmod n$, $n = Pm$ (necessary for a Carmichael number; computed in C with 128-bit modular arithmetic when $P \le 2^{40}$, and otherwise deferred to the Python verifier below); (iv) for each remaining $m$, recover $r$ as an integer root of $dx^3 - (d-2)x - 2n$ for $d = 1, \dots, 2p_1$ (multiply the first equation of (1) by $r$: $2n = dr^3 - (d-2)r$; Theorem 3.1(b) gives $d < 2p_1$), set $q = m/r$, and check all four conditions of Lemma 2.1 together with the primality of $q$ and $r$. Independently of the C program, a Python program using exact integer arithmetic and SymPy factorisation re-tests every survivor of step (ii) that it is given.

**Proposition 3.7** (completeness of the search)**.** *Let $n = p_1 p_2 q r$ be a Williams number with four prime factors, and suppose $n \le B$ (campaign 1) or $p_1 \le X$ (campaign 2). Then the algorithm above visits the pair $(p_1, p_2)$, the candidate $m = qr$ survives step (ii) and step (iii), and step (iv) recovers $(q, r)$ and reports $n$.*

*Proof.* By Theorem 3.1(a) both $p_1, p_2$ are primes $\equiv 1 \pmod{12}$. If $n \le B$ then $p_1^4 < p_1 p_2 q r = n \le B$ and $p_1 p_2^3 < n \le B$, so $p_1 < B^{1/4}$ and $p_2 < (B/p_1)^{1/3}$; in either campaign $p_2 \le \sqrt{24 p_1^3 + 1}$ by Theorem 3.1(c). Hence the pair is in the range scanned. The congruences of Lemma 3.5 hold for $m$, so the pair passes step (i) and $m \equiv m_0 \pmod{\Lambda}$; by Theorem 3.1(b), $p_2^2 < m < 4P^2$, and $m \le B/P$ when $n \le B$, so $m$ is one of the candidates walked in step (ii). By Lemma 3.6 the pair $(m, de)$ is admissible, so if $m \le M^*$ then $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$ is admissible and $m$ is kept, while if $m > M^*$ then $j = de$ lies in the scanned $j$-range and $m$ lies in the exact window for that $j$, so again $m$ is kept. Since $n$ is a Carmichael number, $2^{n-1} \equiv 1 \pmod{n}$ and $m$ survives step (iii) (or is deferred to the Python verifier, which applies the same test). Finally $2n = dr^3 - (d-2)r$ with $1 \le d < 2p_1$ (Theorem 3.1(b)), so the loop over $d$ in step (iv) reaches $d$, the integer cube root recovers $r$, $q = m/r$, and all checks pass. The actual runs are certified by the record described in §6: every block of pairs logged exactly once with contiguous $p_1$ ranges, and every window survivor re-tested in Python. □

Validation. The implementation was tested against brute force on a *relaxed* problem that does have solutions: replace $L_p$ by $(p^2 - 1)/\kappa$ for $\kappa \in \{24, 120\}$. Theorem 3.1(b) and Lemmas 3.5–3.6, which are all the algorithm uses, go through with 2 replaced by $\kappa$ (Theorem 3.1(c) and the Fermat filter are specific to $\kappa = 2$ and are switched off in the relaxed runs, so step (iii) is validated only by the Python re-verification). On the relaxed problem the enumerator finds exactly the solution set that a direct four-fold loop finds (for $\kappa = 24$ the unique solution below $10^{12}$ is $13685 = 5 \cdot 7 \cdot 17 \cdot 23$, for which $d = 27$, $e = 67$, $s = 977$ and $des \cdot \Pi = 24^3 \cdot 5^3$ exactly; the outputs of both programs to $10^{12}$ are archived in `clc/k4/validation/`). This is a regression test of the implementation on an instance with known answer, not an independent validation of the production search, which rests on the completeness argument above and the certification record of §6. Separate self-tests brute-force the two-prime sub-problem $(q, r)$ for small $(p_1, p_2)$ in four relaxed variants (43 solutions) and check every inequality used above, and brute-force the residue class of Lemma 3.5 for 13,037 pairs. On the real problem the enumerator reproduces the empty result below $10^{24}$ of Theorem 4.1 in under ten seconds. One design error was caught by these tests before any production run: the true $j = de$ is *not* in general $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$ (only the filter may use those two values), so the reconstruction step (iv) must not depend on $j$.

**Theorem 3.8.** *There is no Williams number with exactly four prime factors and $n \le 10^{30}$, and there is none at all whose smallest prime factor is at most* 500,000*. Consequently a four-prime Williams number, if one exists, has $p_1 >$ 500,000 and $n > 10^{30}$. (McIntosh and Dipra* [18] *had $p_1 > 2 \times 10^4$ and $n > 10^{24}$.)*

*Computation.* Table 1 summarises the two campaigns (Apple M4 Pro, 14 threads). In each, the single survivor of the Fermat filter is the Carmichael number $26921089 = 13 \cdot 37 \cdot 97 \cdot 577$, for which $38 \nmid 26921090$. Both campaigns were then rerun with every window survivor written out, and certified by a script with hard assertions (`clc/k4/certify_k4.py`): every block of pairs logged exactly once, block ranges contiguous over the admissible primes, per-block counts equal to the production run, and all 1,904 + 1,185 distinct window survivors re-tested in Python (base-2 Fermat test, then factorisation and the four divisibilities on the original tuple); exactly one survivor in each campaign passes the Fermat test, the Carmichael number above, and none is a Williams number. The certification records, with hashes of the binary, source and logs, are `CERTIFY_n1e30.txt` and `CERTIFY_p1_500000.txt`. The second statement combines the second campaign with Corollary 3.3. □

| campaign | pairs $(p_1, p_2)$ | window survivors | Fermat survivors | solutions | wall time |
|---|---|---|---|---|---|
| all $n \le 10^{30}$ | 230,612,582,193 | 1,904 | 1 | 0 | 87 min |
| all $n$, $p_1 \le 500{,}000$ | 86,517,004,140 | 1,185 | 1 | 0 | 146 min |

TABLE 1. The four-prime enumeration (Theorem 3.8).

## 4. No Williams number below $10^{24}$

**Theorem 4.1.** *No Williams number is smaller than* $10^{24}$*. Among the* 308,279,939 *Carmichael numbers below* $10^{24}$*, the largest number of prime factors* $p$ *with* $(p+1) \mid (n+1)$ *is four, first attained at*

$$155003096905548657580 09 = 37 \cdot 103 \cdot 109 \cdot 127 \cdot 307 \cdot 523 \cdot 613 \cdot 1429 \cdot 2089$$

*(the primes* $37, 109, 1429, 2089$*);* 826 *numbers have exactly three. Below* $10^{22}$ *the maximum is three, attained by* 179 *numbers. (Our scan recorded the first number attaining the maximum and the count of three-match numbers; the number of four-match numbers was not recorded and would require a rerun.)*

This is, as far as we have found, the largest bound stated in print for Williams numbers with an arbitrary number of prime factors: Pinch's $10^{16}$ [20] is restricted to an odd number of prime factors; McIntosh and Dipra [18] give $10^{24}$ for four prime factors (indeed for Carmichael numbers with four prime factors whose largest prime $s$ satisfies $(s+1) \mid (N+1)$), note that Feitsma's list of base-2 pseudoprimes below $2^{64}$ contains no Williams number, and report a Pinch-style search for $4 \le d \le 13$ prime factors "for various values of $X$ between $10^{20}$ and $10^{35}$" without stating the bound reached for each $d$.

Method. Carmichael numbers have been tabulated completely to $10^{21}$ by Pinch [20, 21], to $10^{22}$ by Goutier [12] and to $10^{24}$ by Shallue and Webster [23, 24], whose list is distributed by Stamm-Wilbrandt as a PARI/GP binary vector [25]. A Williams number is in particular Carmichael, so it suffices to test every tabulated Carmichael number for the Lucas–Carmichael property. We did this ourselves rather than rely on flags computed by the tabulators: each $n$ was factored (the Miller–Rabin square-root-of-unity splitting, which always succeeds on Carmichael numbers, with a general factoring fallback), the factorisation was checked to be squarefree with product $n$ and to satisfy Korselt's criterion, and the number of primes $p$ with $(p+1) \mid (n+1)$ was recorded. Processing is line-based so that no entry can be skipped at a buffer boundary. Three completeness checks were applied to the $10^{24}$ list: the counts below $10^{22}$, $10^{23}$ and $10^{24}$ (49,679,870; 123,381,982; 308,279,939) equal the published ones [24], the list is strictly increasing, and it begins $561, 1105, 1729, 2465, 2821$. Our per-$\omega$ tallies agree with independently published counts: 1,494,738 three-prime Carmichael numbers below $10^{24}$ (OEIS A132195, [8]) and 6,529,921 eleven-prime ones (OEIS A299711); below $10^{22}$ all eleven per-$\omega$ counts agree with Goutier's tally, and the 179 three-match numbers below $10^{22}$ agree with OEIS A329948 (Suteu; b-file by Eldar). The converse direction was also checked: none of the 129,125 Lucas–Carmichael numbers below $10^{15}$ tabulated by Suteu (OEIS A006972) is Carmichael. Table 2 gives the distribution below $10^{24}$.

**Corollary 4.2.** (a) *A counterexample to Agrawal's conjecture of the Lenstra–Pomerance type (a Williams number all of whose primes are* $\equiv 3 \pmod{80}$*) exceeds* $10^{24}$*.*

(b) *A Baillie–PSW counterexample of the Pomerance type (a Carmichael number with* $(p+1) \mid (n+1)$ *for all* $p \mid n$*) exceeds* $10^{24}$*.*

| $\omega(n)$ | count | $\omega(n)$ | count | $\omega(n)$ | count |
|---|---|---|---|---|---|
| 3 | 1,494,738 | 7 | 62,338,997 | 11 | 6,529,921 |
| 4 | 548,265 | 8 | 99,469,866 | 12 | 547,590 |
| 5 | 3,429,390 | 9 | 81,041,495 | 13 | 15,328 |
| 6 | 19,665,194 | 10 | 33,199,094 | 14 | 61 |

TABLE 2. Carmichael numbers below $10^{24}$ by number of prime factors, as recounted by our scan (total 308,279,939).

Statement (b) adds little to the known absence of Baillie–PSW counterexamples below $2^{64} \approx 1.8 \times 10^{19}$ [11], but it is a different kind of statement: it bounds every construction that proceeds through the Pomerance conditions, not only the tested range.

*Remark* 4.3 (Howe's order-2 Carmichael numbers). By Howe's criterion [14], a squarefree $n$ is a Carmichael number of order 2 iff for each $p \mid n$, $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$. For a Williams number the first option is impossible ($n \equiv p \pmod{L_p}$ and $n \equiv 1 \pmod{p^2-1}$ would give $L_p \mid p-1$), so a Williams number is of order 2 iff $n \equiv p \pmod{p^2-1}$ for all $p \mid n$, i.e. iff $(n-p)/L_p$ is even for every $p \mid n$. Tabulations of order-2 Carmichael numbers therefore bound one 2-adic type of Williams number; we have not pursued this.

## 5. Heuristics

This section is speculative: it records what a simple probabilistic model predicts, as motivation and as a consistency check against the computations. It proves nothing and is not used elsewhere in the paper.

Four prime factors. Condition on Theorem 3.1(a). For $p \equiv 1 \pmod{12}$ the condition $n/p \equiv 1 \pmod{L_p}$ is automatic modulo 12, so model the remaining condition as an event of probability $24/p^2$, independently for the four primes. Summing $\prod_i 24/p_i^2$ over quadruples of primes $\equiv 1 \pmod{12}$ with $p_2^2 \le 24p_1^3 + 1$ and $p_2 < q < r < 2p_1p_2$ (Theorem 3.1) gives an expected count of about $6.3 \times 10^{-7}$ (script `agrawal/k4_heuristic.py`), of which about 96% comes from $p_1 = 13$ (where $n < 1.1 \times 10^{11}$, long excluded by Theorem 4.1); the contribution of $p_1 > X$ decays like $X^{-4}$ (up to logarithms) and is of order $10^{-25}$ for $X = 500{,}000$. On this model the four-prime case is empty with overwhelming probability, and Theorem 3.8 confirms the part of the prediction that can be tested.

Onset of Williams numbers of any size. A cruder Erdős–Pomerance-style model for arbitrary numbers of prime factors is described in Appendix B; it suggests that the first Williams number, if any, has hundreds of digits and on the order of a hundred prime factors.

## 6. Reproducibility

All code is in the repository `https://github.com/razv2025/carmichael-lucas-carmichael`, tagged for this version. The $10^{24}$ scan (`clc/`) consists of the download–decode–verify–scan pipeline, the completeness report, the result files with per-$\omega$ and per-match counts, and SHA-256 checksums of the input data. The four-prime enumeration (`clc/k4/`) consists of the C program, its brute-force reference, the Python re-verifier, the per-block logs of both campaigns (one JSON line per block, so the coverage can be audited), the survivor dumps and the certification records `CERTIFY_*.txt`, and `make test`, which runs the self-tests and the relaxed end-to-end comparison in under a minute. The family searches and the heuristic model are in `agrawal/`, with a PYTEST suite that includes symbolic checks of the identities used in the proofs

of Theorems 3.1 and A.1–A.3. The consultation history, including the errors caught along the way, is in `docs/`.

## 7. Open problems

(1) Is $k = 4$ impossible? By Theorem 3.1, fixing $p_1$ leaves a finite search over $(p_2, q, r)$; the nonzero integer $\delta = des - 8p_1^3 = 8p_1^3(1/\Pi - 1)$ ($\delta \geq 1$ when $\Pi < 1$, $\delta \leq -1$ when $\Pi > 1$) ties the sizes of $p_2, q, r$ to $p_1$ but does not by itself exclude them; McIntosh and Dipra prove $\delta \geq 1$ [18, proof of Thm. 2].
(2) The $k = 5$ case, where both classes modulo 4 are allowed and the analogue of (2) has four factors.
(3) Closing the $m = 4$ polynomial-family stratum with one repeated root.
(4) A rigorous two-sided analogue of Pomerance's upper bound for Carmichael numbers.

## Appendix A. Obstructions to the classical constructions

Carmichael numbers are constructed in two ways: by polynomial families in the manner of Chernick [7], $n(k) = (6k+1)(12k+1)(18k+1)$, where the divisibilities are polynomial identities; and by subset products in the manner of Erdős and Alford–Granville–Pomerance [10, 3, 15], where many primes $p$ with $p - 1$ dividing a fixed smooth $\Lambda$ are harvested and a subset with product $\equiv 1 \pmod{\Lambda}$ is sought. We show that the two-sided versions of both break down in their simplest forms.

A.1. **Polynomial families.** Consider $n(k) = \prod_{i=1}^{m}(a_i k + b_i)$ with integers $a_i > 0$, the linear factors pairwise non-proportional, and require the CLC divisibilities as polynomial identities: $(a_i k + b_i - 1) \mid (n(k) - 1)$ and $(a_i k + b_i + 1) \mid (n(k) + 1)$ in $\mathbb{Q}[k]$. Since a linear polynomial divides $f$ iff $f$ vanishes at its root, this is $n\big((1 - b_i)/a_i\big) = 1$ and $n\big((-1 - b_i)/a_i\big) = -1$. With $D_{ij} = a_i b_j - a_j b_i$ one computes $a_j \frac{1-b_i}{a_i} + b_j = (a_j + D_{ij})/a_i$, so the conditions read, for every $i$,

$$\prod_{j \neq i}(a_j + D_{ij}) = a_i^{m-1} = \prod_{j \neq i}(D_{ij} - a_j). \tag{5}$$

**Theorem A.1** (three factors)**.** *No such family exists for $m = 3$.*

*Proof.* Subtracting the two products in (5) for $i = 1, 2, 3$ gives $a_2 D_{13} + a_3 D_{12} = 0$, $a_1 D_{23} + a_3 D_{21} = 0$, $a_1 D_{32} + a_2 D_{31} = 0$, i.e. $a_2 D_{13} = -a_3 D_{12}$, $a_1 D_{23} = a_3 D_{12}$, $a_1 D_{23} = -a_2 D_{13}$. The identity $a_1 D_{23} - a_2 D_{13} + a_3 D_{12} = 0$ (it is $a \wedge b \wedge a = 0$) becomes $3a_3 D_{12} = 0$, so $D_{12} = 0$ and then $D_{13} = D_{23} = 0$: $b$ is proportional to $a$ and the factors are proportional, contrary to hypothesis. □

**Theorem A.2** (uniform offsets)**.** *No such family exists with $b_i = -1$ for all $i$, nor with $b_i = +1$ for all $i$, for any $m \geq 2$.*

*Proof.* If all $b_i = -1$ then $D_{ij} = a_j - a_i$, the second product in (5) is $(-a_i)^{m-1}$, and the first becomes $\prod_{j \neq i}(2a_j - a_i) = a_i^{m-1}$. The $a_i$ are distinct (non-proportionality). Take $i$ with $a_i$ maximal: for $j \neq i$, $0 < a_j < a_i$ gives $|2a_j - a_i| < a_i$, so the product has absolute value $< a_i^{m-1}$. The case $b_i = +1$ is the mirror image with $\prod_{j \neq i}(a_i - 2a_j)$. □

**Theorem A.3** (generic position)**.** *Put $s_i = (b_i - 1)/a_i$ and $t_i = (b_i + 1)/a_i$. If the $s_i$ are pairwise distinct and the $t_i$ are pairwise distinct, no such family exists, for any $m \geq 2$.*

*Proof.* Let $G(\tau) = n(-\tau)$, of degree $m$ with leading coefficient $c = (-1)^m \prod a_i$. The hypotheses say $G(s_i) = 1$ and $G(t_i) = -1$. With $m$ distinct roots each, $G - 1 = c\prod(\tau - s_i)$ and $G + 1 = c\prod(\tau - t_i)$. Subtracting, $2 = c\big[\prod(\tau - t_i) - \prod(\tau - s_i)\big]$ identically in $\tau$. The coefficient of $\tau^{m-1}$ on the right is $-c\sum(t_i - s_i) = -c\sum 2/a_i \neq 0$, while on the left it is 0. □

**Corollary A.4.** *Every CLC polynomial family has two indices $i \neq j$ with $p_j - 1 = \frac{a_j}{a_i}(p_i - 1)$ or $p_j + 1 = \frac{a_j}{a_i}(p_i + 1)$ as polynomial identities.*

The degenerate strata left open by Theorem A.3 were searched exhaustively using the fact that each factor in (5) divides $a_i^{m-1}$: no family exists with $m = 4$ and all $a_i \leq 500$, with $m = 5$ and $a_i \leq 150$, or with $m = 6$ and $a_i \leq 60$ (45 min, 1.8 h and 54 h respectively). Theorem A.2 was also confirmed by search for $m = 5, 7, 9, 11$ up to $a_i \leq 20000, 5000, 3000, 1500$.

*Remark* A.5. These theorems close families in which the divisibilities hold identically in $k$. They say nothing about families that are CLC only for $k$ in some residue classes, nor about sporadic solutions. Theorem A.1 is the parametric shadow of Proposition 2.3.

A.2. **Two-sided subset products.** In the subset-product method one fixes a smooth modulus $\Lambda$, harvests primes $p$ with $(p-1) \mid \Lambda$, and looks for a subset with product $\equiv 1 \pmod{\Lambda}$; the expected number of solutions is about $2^{(\text{pool size})-\log_2 \varphi(\Lambda)}$, and the method succeeds because the pool grows faster than $\log_2 \varphi(\Lambda)$. For Williams numbers one needs simultaneously $(p+1) \mid \Lambda'$ and product $\equiv -1 \pmod{\Lambda'}$, and the two moduli interact: an odd prime $\ell$ cannot divide both some $p_i - 1$ and some $p_j + 1$ (else $n \equiv 1 \equiv -1 \pmod{\ell}$), so every small prime must be assigned to one side, and the 2-adic branch is fixed. Define the *margin* of a configuration as (number of usable primes) $- \log_2 \varphi(\text{lcm of all required } p \mp 1)$, so that the expected number of two-sided solutions is about $2^{\text{margin}}$.

We measured the margin over three families of configurations: random smooth moduli (3,250 configurations, all with margin $> -14$ searched exhaustively by meet-in-the-middle, no solution); pools of all primes $p \leq X$ with both $p-1$ and $p+1$ $y$-smooth, with the side assignment optimised by hill-climbing (hundreds of restarts); and pools restricted to primes with few prime factors in $p \mp 1$. The best margins were $-8.2$ (tiny three-prime systems), $-29.0/-30.6/-31.0/-36.9$ for $y = 60/100/140/200$, and $-36.9$ for the sparse pools. The wall is flat or worsening in every direction we varied: each usable prime contributes one bit of subset freedom but about 2.5 bits of modulus, because a prime that is useful on one side is almost never useful on the other. This is a measurement, not a theorem. The side-assignment of the small primes was formalised by Zhang [26], who splits the prime factors of the modulus into two groups serving $p-1$ and $p+1$ respectively, conjectures that the number of Williams numbers built from a modulus with $s$ prime factors grows only like $2^{2^{s^{1/2-o(1)}}}$ (against $2^{2^{s(1+o(1))}}$ for Carmichael numbers), and exhibits a set of 1,029 primes expected to contain at least $2^{24}$ Williams numbers, none of which has been found. Our margins are a small-scale measurement consistent with Zhang's conjecture and with the "intractable" judgement of [5].

## Appendix B. A heuristic onset estimate

*This appendix is heuristic.* For comparison, Zhang [26, Remark 7.4] predicts that a Williams number built from his set of 1,029 primes would have about 428 prime factors; the estimate below concerns the smallest Williams numbers of any origin and gives fewer, but likewise many, prime factors. An Erdős–Pomerance-style count of $k$-factor products of primes with $p \mp 1$ both $y$-smooth, including the side-consistency factor $2(1-1/\ell)^k - (1-2/\ell)^k$ for each odd prime $\ell \leq y$ and a factor $2^{1-k}$ for the 2-adic branch, calibrated on the one-sided side against Pinch's counts $C(10^{15}) = 105{,}212$, $C(10^{18}) = 1{,}401{,}644$ and on the two-sided side against our measured counts of twin-smooth primes, places the first Williams number near $10^{208}$–$10^{490}$ with $k \approx 80$–$110$ and $y \approx 40$–$90$, and the first Lenstra–Pomerance counterexample to Agrawal's conjecture near $10^{2400}$–$10^{2900}$ with $k \approx 270$. The ranges reflect the sensitivity to the per-prime calibration; the qualitative conclusion—far beyond exhaustive search, reachable only by a construction that

circumvents §A—is robust. The same machinery places the first Carmichael number near $10^3$ (truth: 561).

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